

Low-Energy Dipole Strength Distribution in ^{102}Ru : Coexistence of Scissors and Octupole Resonances

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The dipole response of ^{102}Ru has been investigated for the first time using nuclear resonance fluorescence, providing comprehensive and detailed information about the nucleus. The $M1$ strength, concentrated in the 3–4 MeV region, is consistent with a scissors resonance. As a surprising new feature, a pronounced $E1$ resonancelike structure at 4.5–5.5 MeV has been found. It is interpreted as an octupole resonance and provides evidence for low-spin octupole correlations in this excitation energy regime. The experimental results are compared with quasiparticle random phase approximation calculations, which reproduce the gross features of the cumulative strength. These observations support the coexistence of $M1$ scissors and octupole-induced $E1$ resonances as a general feature of deformed nuclei with octupole softness.

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Low-energy dipole excitations in deformed and transitional nuclei provide a sensitive probe of the interplay between single-particle structure and collective motion. Whereas the dipole response at excitation energies well above the particle threshold is dominated by the isovector giant dipole resonance (IVGDR), the subthreshold region often exhibits a much richer admixture of electric and magnetic modes whose microscopic origin remains a subject of active investigations [1–9]. In particular, a central challenge is to distinguish coexisting $E1$ and $M1$ excitations experimentally and to establish the physical origin of enhanced low-energy $E1$ strength, especially in systems where deformation, shell structure, and multiphonon correlations may all contribute within the same energy interval [10–17].

The distribution of subthreshold $E1$ strength has attracted sustained interest over the past two decades owing to its possible connection to several distinct structural mechanisms, including neutron skin effects, multiphonon excitations, and octupole correlations. The pygmy dipole resonance (PDR) is generally understood as an enhancement of the $E1$ strength function relative to the low-energy tail of the IVGDR. In β -stable nuclei, the PDR is typically concentrated around 8 MeV, whereas in weakly bound neutron-rich systems it may be shifted to lower excitation energies, reaching about 5 MeV [13–16,18]. It is often interpreted in terms of an oscillation of the neutron skin against the isospin symmetric core [13,14,16]. However, the extent to which this intuitive interpretation is applicable, in general, and, in particular, to β -stable nuclei, remains unclear [15].

In addition to the PDR, low-energy $E1$ strength may also originate from the coupling of quadrupole and octupole shape degrees of freedom [9,19]. In weakly deformed nuclei, the relevant states are interpreted as 1^- excitations formed by coupling the 2_1^+ quadrupole and 3_1^- octupole phonons, with the dipole operator augmented by a two-phonon term that accounts for the associated modification of the charge density. In well-deformed nuclei, the reflection-asymmetric Coulomb field generated by a quadrupole-octupole-deformed charge distribution induces an intrinsic dipole moment and thereby enhances low-energy $E1$

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strength [20]. Most nuclei are expected to exhibit structures intermediate between these two limits. In the present case, the relevant picture is that of an octupole phonon built on a well-quadrupole-deformed nucleus. Beyond their structural significance, the associated enhancement of low-energy $E1$ strength can affect radiative neutron-capture rates relevant to stellar nucleosynthesis [21–24].

The low-energy $M1$ response likewise comprises two distinct modes. At higher excitation energies, it is dominated by the isovector spin-flip (SF) resonance, which corresponds to the reorientation of nucleon spins relative to one another. The SF mode is related to the $T = 0$ component of the isovector Gamow-Teller resonance [25] and generated by transitions between spin-orbit partners at energies of the order of the spin-orbit splitting. The $M1$ absorption associated with this mode has been extensively reviewed in Refs. [6,26]. At lower energies, the scissors mode (SM) has been observed systematically in rare-earth and actinide nuclei [26]. It is understood as a deformation-dependent isovector oscillation of proton and neutron currents against each other. Its strength and fragmentation pattern provide important information on the interplay between deformation, shell structure, and proton-neutron coupling. The $M1$ transitions also contribute significantly to the low-energy γ -ray strength function and are, therefore, relevant to the same applications discussed above [27,28].

Disentangling the $E1$ and $M1$ components of the low-energy dipole strength distributions experimentally remains challenging due to the inherent limitations of conventional photon-scattering techniques. Most of the existing information on $E1$ and $M1$ distributions below the neutron separation energy has been obtained from nuclear resonance fluorescence (NRF) experiments using bremsstrahlung beams [9,15]. While such measurements provide high-resolution data, they lack the sensitivity required to unambiguously determine the parity of individual dipole states. Consequently, the separation of $M1$ and $E1$ components relies on model-dependent assumptions and systematics, which limit the reliability of extracted strength distributions. In contrast, NRF measurements performed with quasimonoenergetic, linearly polarized photon beams enable model-independent parity assignments based on angular distribution asymmetries, thereby providing a clean separation of $M1$ and $E1$ excitations [29,30]. Such capabilities, currently available at the High Intensity Gamma-ray Source (HI γ S) facility [30,31] of the Triangle Universities Nuclear Laboratory, are essential for detailed studies of the low-energy dipole strength distribution [32].

This Letter presents the first NRF study of ^{102}Ru carried out at HI γ S. The use of polarized photons enables, for the first time in this mass region, a clean and model-independent determination of the parity of dipole states. This crucial advance allows the $M1$ and $E1$ components of the dipole response to be distinguished with high confidence, thereby enabling a detailed characterization of the dipole strength

distribution at low energies. The results reveal three principal features: (i) a concentration of $M1$ strength in the 3–4 MeV region consistent with a scissors resonance, (ii) a strong $E1$ transition at 2.95 MeV consistent with quadrupole-octupole two-phonon excitation, and (iii) a pronounced resonancelike concentration of $E1$ strength centered around 4.5–5.5 MeV. The latter is interpreted as an octupole resonance. It lies below the energy range typically associated with the PDR in stable nuclei and above the expected energy of the candidate two-phonon structure.

Calculations based on the tilted axis cranking (TAC) variant of the quasiparticle random phase approximation (QRPA) qualitatively reproduce the observed $M1$ scissors resonance. They account for the gross features of the IVGDR and predict an enhancement of $E1$ strength around 10 MeV, where a PDR is expected in β -stable nuclei (see Supplemental Material [33]). However, the experimentally observed concentration of low-energy $E1$ strength near 5 MeV is not naturally accounted for by the pure dipole response alone. Instead, the combined experimental evidence and TAC-QRPA results point to the presence of additional $E1$ strength associated with octupole collectivity. This interpretation is further reinforced by recent experimental evidence for stable octupole deformation in the neighboring nucleus ^{100}Ru [37].

The NRF measurement was performed on 2 g of ^{102}Ru powder isotopically enriched to 99.25(5)%. Further experimental details will be presented in a forthcoming paper and in Ref. [38]. The photon beam was scanned across the energy range 2.30–8.70 MeV, covering the low-lying dipole strength region up to the particle emission threshold. The scan was carried out in increments of about 3%, corresponding to the typical beam energy spread (FWHM), with data collected for approximately three hours at each setting. The emitted γ rays were detected using the clover array setup [39], a hybrid system comprising eight HPGe clover detectors and 12 CeBr $_3$ scintillators positioned upstream in a semispherical geometry around the target. To leverage the parity sensitivity offered by the polarized beam, four HPGe detectors were mounted at a polar angle of $\theta = 90^\circ$ relative to the incident beam direction. Two of these were located in the polarization plane of the beam at azimuthal angles $\phi = 0^\circ$ and 180° , while the other two were placed perpendicular to the polarization plane at $\phi = 90^\circ$ and 270° . This geometry, combined with the linear polarization of the HI γ S beam, is optimal for distinguishing $M1$ from $E1$ transitions, since the angular distribution of scattered photons is orthogonal for the two parities [40]. To improve sensitivity to quadrupole and higher multipolarities, additional HPGe detectors were positioned at backward angles relative to the incident beam direction, at polar angles $\theta \in \{125^\circ, 135^\circ\}$ and azimuthal angles $\phi \in \{45^\circ, 135^\circ\}$. Their angular response also provides discrimination between dipole and quadrupole excitations. Typical angular distribution patterns for excitations induced

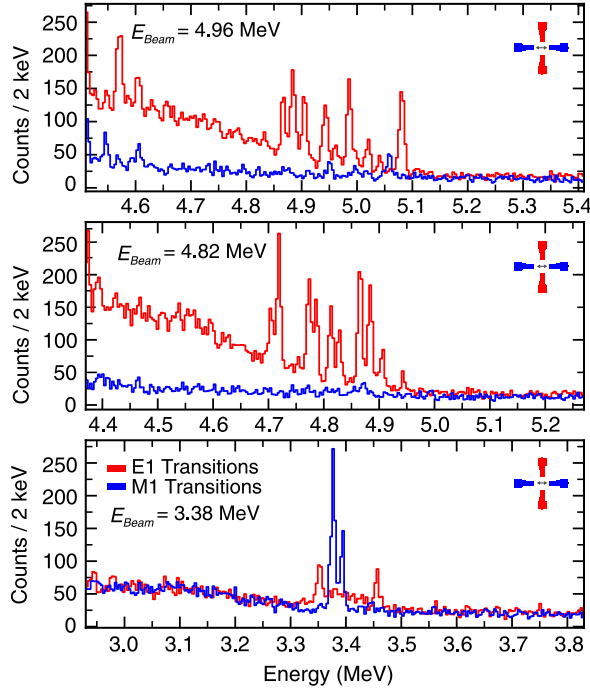


FIG. 1. Primary (elastic) decay of levels excited at bombarding energies of 4.96 (top), 4.82 (middle), and 3.38 MeV (bottom). Red spectra were measured with detectors perpendicular to the polarization plane, and blue spectra with detectors parallel to it. The contrasting angular response distinguishes $E1$ (red) from $M1$ (blue) dipole transitions.

by linearly polarized photons are shown in Fig. 3 in Ref. [40]; further details of the clover array configuration are given in Ref. [38].

Figure 1 displays representative NRF spectra for ^{102}Ru , measured using linearly polarized photon beams with centroid energies of 3.38, 4.82, and 4.96 MeV. Although the full measurement covered the broader range from 2.30 to 8.70 MeV, the present Letter focuses on the subthreshold region where the enhanced $E1$ clustering is observed; an extended spectral presentation will be included in a forthcoming paper. Peaks corresponding to ground-state transitions from excited dipole states were recorded simultaneously in detectors positioned at azimuthal angles parallel and perpendicular to the polarization plane as well as in detectors at backward angles. The parity of each dipole state was determined from the polarization asymmetry measured between the detector groups parallel and perpendicular to the polarization plane [40]. In this geometry, $E1$ transitions exhibit maximum intensity perpendicular to the polarization plane, whereas $M1$ transitions peak in the plane. The extracted asymmetries are plotted as a function of transition energy in the top panel in Fig. 2, and the complete list of level energies, branching ratios, multipolarities, and reduced transition probabilities are provided in Table I in Supplemental Material [33].

The absolute photon flux was determined with a monitoring system, consisting of plastic scintillator paddles

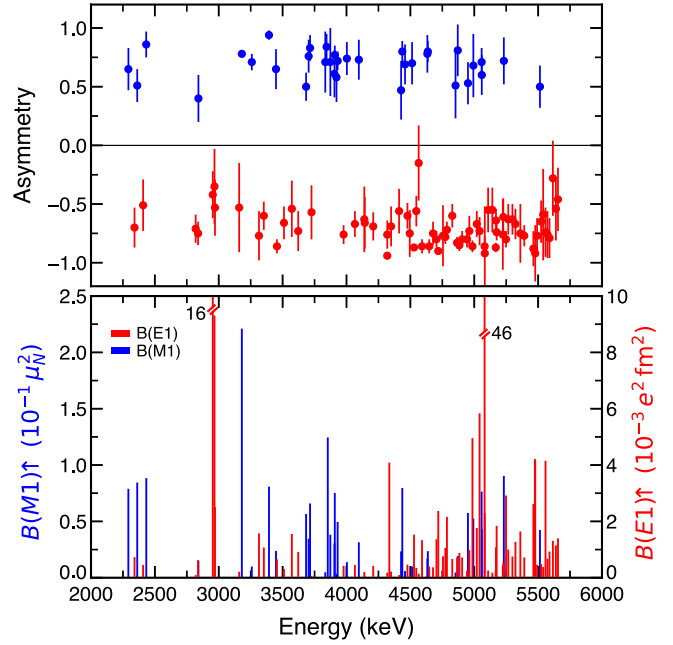


FIG. 2. Polarization asymmetries (top) and reduced transition probabilities (bottom) for dipole transitions in ^{102}Ru . The asymmetries approach limits of $+1$ for magnetic (blue) and -1 for electric (red) transitions. $B(M1)\uparrow$ (blue, left axis) and $B(E1)\uparrow$ (red, right axis) probabilities are plotted as a function of energy; the axes are scaled independently. Transitions at 2953.2 (19) and 5080.9(31) keV are scaled for visibility, with actual values indicated. A clear concentration of $E1$ strength appears in the 4.5–5.5 MeV region, centered around 4.9 MeV.

calibrated against a large NaI(Tl) detector. Details of the calibration procedure are given in Ref. [41]. In addition, for each beam setting, the energy profile was measured with a coaxial HPGe detector placed in the beam at $\theta = 0^\circ$. The measured spectra were corrected for the detector response using the HORST code [42]. The resulting beam profiles, combined with the monitor data, provided the energy-dependent photon flux $\Phi(E)$ required to determine the integrated cross sections. The reduced transition probabilities $B(E1)\uparrow$ and $B(M1)\uparrow$ extracted from the measured cross sections are displayed in the bottom panel in Fig. 2. A concentration of $M1$ strength is observed in the ~ 2 –4 MeV region. The summed $\Sigma B(M1)\uparrow \approx 1\mu_N^2$ strength in this interval is consistent with that expected for scissors resonances in transitional nuclei [26].

The striking feature in Fig. 2 is the observation of a strong $E1$ transition at 2.95 MeV with $B(E1)\uparrow = 0.016 e^2 \text{ fm}^2$ and a resonancelike accumulation of $E1$ strength centered at $E_\gamma \approx 4.9$ MeV with a summed strength of $\Sigma B(E1)\uparrow \approx 0.1 e^2 \text{ fm}^2$. As shown in Fig. 3, these exhaust a sizable fraction of the subthreshold dipole strength. Their $B(E1)$ values exceed those typically associated with low-energy single-particle $E1$ transitions by several orders of magnitude. Such transitions are usually strongly suppressed, with strengths of the order of $10^{-5} - 10^{-6}$ W.u. [43], corresponding in the case of ^{102}Ru

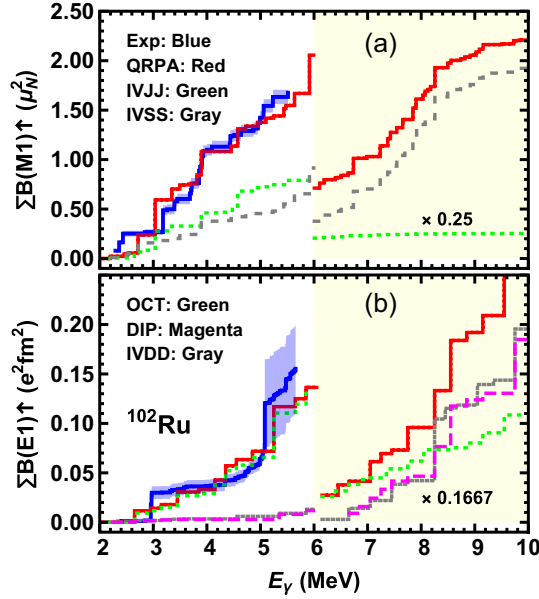


FIG. 3. Experimental cumulative $M1$ (a) and $E1$ (b) strengths, indicated by the blue curves with shaded error bands, in comparison with the corresponding QRPA calculations. In the $M1$ calculations, the red line presents the full contributions in the $M1$ operator (1). The green and gray lines present the IVJJ and IVSS contributions, respectively. For $E_\gamma > 6$ MeV, the calculated results are rescaled by a factor of 0.25. In the $E1$ calculations, the red line shows the full contributions from the $E1$ operator (2). The gray line shows the result without octupole correlation ($\kappa_{\text{IVOO}} = 0$), and the green and magenta lines the separate contributions from the octupole and dipole transitions, respectively. For $E_\gamma > 6$ MeV, the calculated results are rescaled by a factor of 0.17.

to roughly $10^{-5} - 10^{-6} e^2 \text{ fm}^2$; see also Ref. [44] for representative examples. The strong suppression of the $E1$ single-particle transitions at low energies is commonly attributed to their coupling to the IVGDR, with additional reduction in strength arising from the weak overlap between high- j intruder and normal-parity orbitals.

Octupole collectivity provides a natural explanation for the observed large $E1$ strength, as it is known to generate enhanced electric dipole transitions with intrinsic dipole moments of the order of $D_0 \approx 0.3 e\text{fm}$ [20,45], consistent with the observed summed strength, $\Sigma B(E1) \uparrow \approx 0.1 e^2 \text{ fm}^2$. As illustrated in Fig. 24 in Ref. [20], nuclei with neutron numbers $N \approx 56$ are expected to exhibit octupole correlations due to the presence of $\Delta l = 3$ and $\Delta j = 3$ orbitals from the $[d_{5/2}^{-1} h_{11/2}]$ configurations near the Fermi surface. Indeed, recent experimental evidence for stable octupole deformation has been reported in neighboring ^{100}Ru [37]. Moreover, the presence of strong individual $E1$ transitions observed in NRF experiments has long been discussed as signatures of octupole correlations [9]. In weakly deformed nuclei, the lowest such state is commonly interpreted as a two-phonon 1^- excitation resulting from the coupling of the 2_1^+ quadrupole

and 3_1^- octupole phonons. In well-deformed axial nuclei, by contrast, the 1^- state is interpreted as the bandhead of a rotational sequence built on the $K^\pi = 0^-$ octupole vibrational phonon. References [9,20] discuss several theoretical approaches for incorporating octupole correlations into the dipole operator. In the present Letter, the formulation of Ref. [44] is adopted.

The QRPA calculations were performed within the TAC framework using a triaxial deformed mean field [46] and separable isoscalar (IS) and isovector (IV) multipole-multipole residual interactions [47]. The full Hamiltonian, comprising a modified harmonic oscillator mean field, a cranking term, and adjustable IS and IV dipole, spin, angular momentum, and octupole residual interactions, is given in Supplemental Material [33]. The use of simplified interactions with adjustable strengths enables exploration of the role of specific correlations. The coupling constants $\kappa_{\text{ISDD}} = -1 \times 10^3 \text{ MeV/fm}^2$ and $\kappa_{\text{ISJJ}} = -1 \times 10^3 \text{ MeV}/\hbar^2$ were chosen to shift the spurious center-of-mass and reorientation modes [Eq. (2) in Supplemental Material [33]] to high excitation energy, effectively decoupling them from the intrinsic vibrational modes [48]. The remaining coupling constants were adjusted to reproduce the gross properties of the measured response. For the $E1$ calculations, each coupling constant κ was scaled relative to the self-consistent value $\kappa_\lambda^{\text{sc}}$, obtained from the oscillator model [49], using a scaling factor η , such that $\kappa = \eta \kappa_\lambda^{\text{sc}}$. The mean field was determined using the equilibrium deformation of $\epsilon = 0.183$ and $\gamma = 0^\circ$ and pairing gaps of $\Delta_p = 1.04 \text{ MeV}$ and $\Delta_n = 0.97 \text{ MeV}$, which corresponds to 80% of the experimental even-odd mass differences. These parameters provide a satisfactory description of the low-spin rotational structure, as shown in Fig. S1 in Supplemental Material [33].

The QRPA strength function is computed using complex contour integration (see, e.g., Ref. [47]). With a sufficiently narrow smoothing width, the calculation yields discrete peaks corresponding to individual transitions from the ground state, where the area under the peak gives the transition probability $B(E1/M1) \uparrow$. Because QRPA does not reproduce individual transitions one to one, a quantitative comparison with experimental data was achieved using cumulative strengths $\Sigma B(M1/E1) \uparrow$ and spectral densities $S(M1/E1, E) \uparrow$ obtained by averaging the line spectra by means of a normalized Breit-Wigner distribution of width Δ [50].

Figure 3(a) compares the experimental cumulative $\Sigma B(M1) \uparrow$ distribution with the QRPA results obtained with $\kappa_{\text{IVSS}} = -1.0 \text{ MeV}/\hbar^2$ and $\kappa_{\text{IVJJ}} = -0.1 \text{ MeV}/\hbar^2$. In these calculations, the $M1$ transition operator is expressed as [25]

$$\mathcal{O}(M1, \nu) = \sqrt{\frac{3}{4\pi}} \left(\frac{1}{2} J_\nu^{\text{IS}} + \frac{1}{2} J_\nu^{\text{IV}} + g^{\text{IS}} S_\nu^{\text{IS}} + g^{\text{IV}} S_\nu^{\text{IV}} \right). \quad (1)$$

With the free-nucleon gyromagnetic factors g_s^π and g_s^ν attenuated by a factor of 0.7, one obtains the values

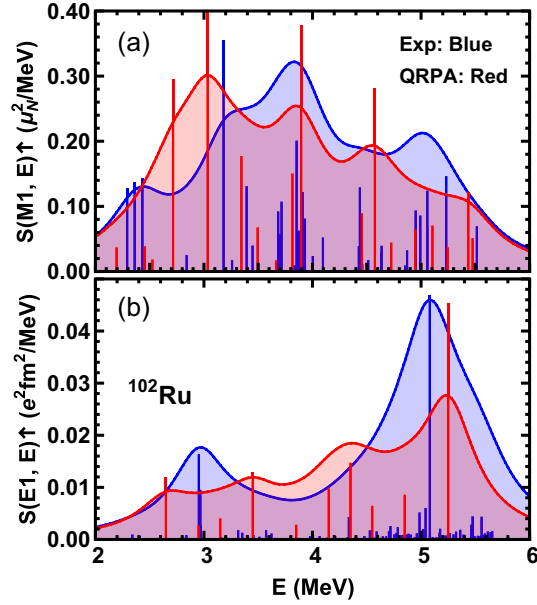


FIG. 4. Experimental and QRPA $M1$ (a) and $E1$ (b) spectral densities (shadows) obtained by smearing the corresponding line spectra (vertical lines) using a Breit-Wigner weighting function with width $\Delta = 0.6$ MeV. The $M1$ and $E1$ line spectra are added in units of $0.625\mu_N^2$ and $e^2 \text{ fm}^2$, respectively.

$g^{\text{IS}} = 0.12\mu_N$ and $g^{\text{IV}} = 2.79\mu_N$. As a consequence, the IS spin contribution S^{IS} is strongly suppressed relative to the S^{IV} term. The operator J^{IS} corresponds to total angular momentum and does not contribute to intrinsic transitions, once the spurious center-of-mass motion is removed. The dominant physical contributions to the $M1$ strength, therefore, arise from the operators J^{IV} (IVJJ) and S^{IV} (IVSS), which represent, respectively, the SM mode, corresponding to the oscillation of the proton angular momentum against the neutron angular momentum, and the SF resonance, associated with the reorientation of nucleon spins relative to one another.

As seen in Fig. 3(a), the experimental cumulative $\Sigma B(M1)\uparrow$ values, corresponding to the blue curve, are qualitatively reproduced by the QRPA calculations (red curve). The total observed strength of $1.7\mu_N^2$ is consistent with the sum rule expectation for a deformation of $\beta = 0.20$ (see Fig. 24 in Ref. [26]). Similarly, the experimental spectral density $S(M1, E)\uparrow$, displayed in Fig. 4(a), exhibits a broad resonance around 3.6 MeV, consistent with the established systematics of the scissor mode as presented in Fig. 27 in Ref. [26]. Relative to the experimental distribution, the QRPA strength is slightly shifted toward lower energies, with a maximum at 3 MeV; nevertheless, the calculation reproduces the observed double humped structure.

To explore the effects of the different components on the underlying structure, the separate contributions of IVJJ (green curve) and IVSS (gray curve) to the $\Sigma B(M1)\uparrow$ distribution are also provided in Fig. 3(a). At lower

energies, the $M1$ strength contains comparable SM and SF contributions, with the SM component becoming progressively important toward larger E_γ values. For $E_\gamma > 6$ MeV, the SM component saturates, while the SF term increases rapidly as the Gamow-Teller resonance region is approached. This evolution is consistent with the expected transition from predominantly scissorslike motion, at low energy, to spin-flip-dominated strength at higher excitation.

In contrast to the $M1$ response, whose overall behavior is reasonably understood in terms of the interplay between the well-established scissors and spin-flip resonances, the origin of the observed low-energy $E1$ strength is structurally less straightforward. The corresponding QRPA results are presented in Fig. 3(b). For this, the coupling constant $\kappa_{\text{IVDD}} = -1.5\kappa_1^{\text{sc}}$ was chosen such that the maximum of the IVGDR peak reproduces the experimental one in the isotope ^{100}Mo , since such data are not yet available for ^{102}Ru . This procedure provides a reasonable description of the IVGDR and yields nearly identical QRPA spectra for the two isotones (see Fig. S2 in Supplemental Material [33]), thus justifying the adopted value of κ_{IVDD} . As indicated in Fig. 3(b), the cumulative $\Sigma B(E1)\uparrow$ strength obtained with only the IVDD interaction (gray curve) remains below $0.01 e^2 \text{ fm}^2$ over the experimental energy range, substantially underpredicting the experimental distribution (blue curve). The $B(E1)\uparrow$ values fluctuate around $10^{-3} e^2 \text{ fm}^2$, consistent with the fact that the small overlap of the single-particle wave functions causes a reduction by a factor of 10 in addition to the screening by the IVGDR.

The observed surplus of $0.03\text{--}0.16 e^2 \text{ fm}^2$ over the IVGDR tail is attributed to the presence of octupole correlations in a quadrupole-deformed nucleus. References [44,51] treated such an enhancement phenomenologically by introducing the augmented intrinsic $E1$ transition operator

$$\mathcal{O}(E1, \nu) = e_{\text{eff}} r Y_{1\nu} + e_{\text{eff}} b_\nu r^3 Y_{3\nu}, \quad (2)$$

where the first term represents the dipole operator, partially screened by the IVGDR, while the second accounts for the dipole moment induced by the octupole phonon on the charge quadrupole, with the parameters b_ν determining the polarizability. In that framework [44,51], the formalism was applied to $E1$ transitions between one-quasiparticle states in odd- A rare-earth nuclei, where the dipole term alone was found to underestimate the experimental $B(E1)$ values significantly. A satisfactory description was obtained only after incorporating the octupole QRPA results of Ref. [52] and allowing the coupling parameters b_0 and $b_{-1} = b_1$ to vary for each rotational band.

In the present Letter, a modified version of Eq. (2) is employed. The bare charges $e_{\text{eff}}^p = e$ and $e_{\text{eff}}^n = 0$ are used, since the screening by the IVGDR is already taken into account through the IVDD interaction with

$\kappa_{\text{IVDD}} = -1.5\kappa_1^{\text{sc}}$. A single overall parameter $b_\nu = b$ is introduced and determined together with the octupole-octupole (OO) interaction strength by fitting the calculated cumulative $\Sigma B(E1)\uparrow$ values to the experimental distribution. The best fit, represented by the red curve in Fig. 3(b), is obtained for $\kappa_{\text{IVOO}} = 4.0\kappa_3^{\text{sc}}$ and $b = 0.009 \text{ fm}^{-2}$. The value of b is of the same order of magnitude as those reported in Refs. [44,51]. The decomposition of the calculated strength in Fig. 3(b) indicates that the dipole term, represented by the magenta curve, provides only a weak background, while the green curve, corresponding to the octupole contribution, accounts for most of the additional strength between about 4 and 6 MeV. Their combined contributions reproduce the experimental $\Sigma B(E1)\uparrow$ strength well.

This observation is consistent with recent self-consistent Hartree-Fock-Bogoliubov + QRPA studies of deformed nuclei [53], which indicate that axial deformation can induce dipole-octupole coupling through the $K^\pi = 0^-$ and 1^- components and thereby enhance the low-energy dipole response. This interpretation is further supported by the full QRPA calculations, which reproduce the pronounced peak in the experimental spectral density $S(E1, E)\uparrow$ at $E_\gamma \approx 5 \text{ MeV}$, although not the isolated strong transition near 2.95 MeV [see Fig. 4(b)]. The latter may instead arise from a quadrupole-octupole, two-phonon-like excitation, expected near $E(2_1^+) + E(3_1^-) = 2518 \text{ keV}$, which is close to the observed line. Such a structure is not expected to be described within the present TAC-QRPA framework, which can account only for the gross magnitude and localization of the observed low-energy $E1$ enhancement. The concentration of $E1$ strength at $E_\gamma \approx 5 \text{ MeV}$ is, therefore, interpreted as a low-lying octupole resonance in the dipole absorption spectrum, generated by deformation-induced dipole-octupole mixing.

In conclusion, the electromagnetic dipole response of ^{102}Ru has been investigated using monochromatic linearly polarized photon beams. The parity-tagged (γ, γ') data establish a concentration of $M1$ strength in the 3–4 MeV region consistent with a scissors resonance. In addition, a localized concentration of $E1$ strength centered near 5 MeV reveals an unexpected enhancement of the low-energy dipole response above the expected dipole background from the GDR. Taken together with the QRPA analysis and the known octupole softness in this mass region, the data support an interpretation of this structure as a low-lying octupole resonance generated by octupole-induced dipole strength. The present results, therefore, establish the coexistence of an $M1$ scissors mode and a low-energy octupole resonance in ^{102}Ru and identify this nucleus as a benchmark for the interplay of spin, quadrupole, and octupole correlations. They also motivate corresponding parity-resolved studies in neighboring transitional nuclei to clarify the broader systematics of octupole-driven dipole excitations below the particle-emission threshold.

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Data availability—The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

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